Social Finance

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Abstract

We propose a framework for understanding how social investors' who seek to maximize a combination of private and social returns from investments in a portfolio of new and established microfinance institutions. The model takes into account the endogeneity of loan contract terms as well as the capital structure of the financial institutions that may emerge to serve target groups of borrowers differentiated primarily by their levels of initial average net worth, and how social investments might transform those patterns. We build upon one of the workhorse models of modern corporate finance (Tirole, 2006) which features limited liability, multiple layers of moral hazard and costly monitoring to explain patterns of financial intermediation, and in our framework, the role and modes of social investment. We pinpoint the role of the subsidies and guarantees implicit in social investors' equity and quasi-equity investments and the role they play in attracting private capital investors and sustaining productivity-enhancing financial intermediation that might otherwise not have taken place.

1 Introduction

The working modes of philanthropy and business are blurring in two ways. First, philanthropists are increasingly using methods borrowed from for-profit investors. The strategies of venture capitalists are being adapted by "venture philanthropists" seeking to fund socially-minded institutions with promising ideas at early stages of development. They hope that the ideas will form the basis of sustainable institutions capable of serving populations at wide scale. As a recent survey of philanthropy put it:

> The modern philanthropist dislikes the term charity, preferring to speak about his social investments. His language is entrepreneurial, sprinkled with references to metrics, scalability, leverage, and venture philanthropy (Weisberg, 2006).

The changes are expanding to foundations with the increasing favor of program-related investments over grants (PRIs include loans, equity shares, recoverable grants, linked deposits, and guarantees); parallel changes are afoot within government agencies (Marble, 1989).

Oikocredit, for example, a Dutch social investment organization, has since 1978 been making long-term loans to "enterprises and institutions for the empowerment of poor people in developing countries ." Their first "social investment" in microfinance, for example, was a $150,000 loan for 5 years at 9% interest to Alternativa, a non-governmental organization providing microcredit to street vendors and small-scale entrepreneurs in Lima, Peru. Between 1994 and the end of 2004, Oikocredit made a cumulative total of 285 loans or investment agreements to 161 institutions in 41 countries, totaling $88 million. Their net profit, returned to shareholders as a dividend, was 2%. The dividend allowed investors a modest return on capital, but they earned substantially less than alternative investments. Their relative loss is their implicit charitable contribution (Heinen, 2005).

The second turn is that many socially-minded institutions receiving funding also increasingly look like businesses. Museums have raised entry prices and run ambitious shops with mail-order catalogues. Non-profit schools are hiking tuitions. Environmental groups buy and sell land. Community development programs work with banks to facilitate mortgages in low-income neighborhoods. Indeed, some of these institutions are forming commercially viable entities and competing in the marketplace on the strength of their balance sheets alone. Opportunity International, a US-based network of microfinance institutions, owns a for-profit insurance company. BRAC, the largest NGO in Bangladesh, owns a for-profit bank. Chicago's Shorebank, the largest community development bank in the United States, is a holding company composed of non-profit and for-profit entities. BASIX has adapted the Shorebank model to South India.

The microfinance sector is home to some of the best-known examples of the new hybrids, and is in many ways the leading edge. Muhammad Yunus, the 2006 Nobel Peace Prize winner and founder of the Grameen Bank, a financial institution for the poor of Bangladesh, created a "social business" funded by social investors.

We tackle the logic and tensions inherent in social finance—the support, with philanthropic objectives, of nonprofits, social businesses like the Grameen Bank, and profit-maximizing businesses serving the poor. The principles behind the new world of philanthropy and social action have not been well-explored by economists. We develop a theory of "social finance" to parallel the modern theory of corporate finance.

Most generally, we assess the justification for the subsidy inherent in social investment. We go further to describe a framework that describes how subsidy can be deployed for optimal impact. A focus is on using funds to build institutions that can achieve scale and attract additional revenue sources and investments. Leverage is critical, and we describe two important kinds of leverage. One, having to do with mobilizing locally-held capital, is often overlooked and is highlighted below.

Our consideration of effective social investment begins with two ideas. The first is that effective giving can lead to multiplier effects. It can "crowd in" other resources by catalyzing the investments of others, increasing the recipient's ability to "leverage" its resources. Leverage (i.e., to be able to obtain funds to use in addition to an institution's own equity) is critical as it determines an institution's ability to reach wide scale. Effective social investment can be used to expand leverage and hence market coverage, holding constant other social concerns.

The second consideration is that philanthropic giving can change the incentives for recipients to work hard (as can commercial lending).

In this framework, profitability is neither necessary nor sufficient for attracting commercial capital. Contrary to some popular claims, earning profits – even high profits –will frequently not expand the leverage of microfinance to take those institutions to wide scale. The presence of a social investor can however tip the scales for private resources to pour in. Total output may be enhanced by supporting institutions that would not exist if profitability was the sole concern. Of course it is no great surprise that a subsidized investment can expand an activity or a sector, the claim however is that social investors and philanthropic capital can expand output and welfare by repairing and expanding markets.

The framework has a specific implication for the most effective use of social investment. We describe arguments that push social investment either "high" or "low" when directed to poor communities. Specifically, we show that smart subsidy involves allocations that either go to institutions serving relatively well-off (but still poor) customers, where the potential for leverage is maximized, or to institutions serving the poorest customers. There is less "bang for the buck" in supporting a middle range of institutions that are self-sufficient with their own resources but for which leverage (and thus potential scale) is limited.

1.1 Two layers of moral hazard

The analysis focuses on social investment in microfinance institutions. The particular gap filled by social investors is created by the unwillingness of most commercial banks to serve the poor. One well-known difficulty for banks stems from moral hazard on the part of borrowers. In practice, though, moral hazard is mitigated by microfinance institutions through the development of technologies, contracts, and management practices that limit risk. All of this takes effort, but the high repayment rates achieved by leading practitioners are testimonies to impressive successes. The essential insight in mitigating moral hazard is that incentives are improved by exposing people to the consequences of their actions. In the microfinance context, the denial of new loans in the case of default (via "progressive lending") and the peer pressure associated with group lending can reduce moral hazard by adding serious repercussions for bad outcomes (Armendariz de Aghion and Morduch, 2005).

Given the success in controlling moral hazard, investors ought to be interested in funding microfinance institutions and expanding their scale. Investments should be pouring in, and they are starting, but most investments are made by social investors with explicit philanthropic objectives.

While social investors are willing to forego the kind of financial returns that alternative investments could bring, they nonetheless work hard to preserve their capital. The result is that most money channeled through microfinance investment funds and via International Financial Institutions supports "top-tier," mature, financially self-sufficient institutions (Abrams and von Stauffenberg, 2007). The growing, not-yet-profitable institutions that are needed to expand the sector are targeted by less than one fifth of the funds in the MicroBanking Bulletin's survey of funds (Barres, 2005:53).

One of the chief difficulties for investors—whether driven by commercial or social ends–is a second layer of moral hazard. The basic moral hazard problem described above involves the microfinance institution and the customer. This second problem involves the investor and the microfinance institution. Once it is the investor's resources that are ultimately at risk (rather than just the microfinance institution's resources or the customers'), the incentives are reduced for the institution to be diligent in managing finances, monitoring loans, and innovating. The incentives to properly screen potential customers weaken, as do the incentives to chase down delinquent borrowers.

The risks are not trivial. Oikocredit, for example, as of late 2004 faced a portfolio at risk (90-day) of 7 percent on its $67 million social investment portfolio (Heinen 2005: 33). In a review of 104 loan contract disbursed before the end of 2002, the average repayment rate was about 91 percent. Seventy four percent of contracts were found to be repaid "almost always on time," but "15 percent require continuous monitoring to follow overdue payments closely, and another 11 percent can be classified as bad-debt cases that are persistent in their default. The latter cases come with high monitoring and legal costs and are likely to lead to (partial) loan loss in the end." The challenges among the African institutions are greater: "Only 57 percent of MFIs in the African sample manage to pay "on time," while more than one out of four are structurally delayed in payment of invoiced amounts (Heinen 2005: 34-35)." The worst performers tended to newly-started, medium-sized urban banks or NGOs in Africa or Asia (Heinen 2005: 39).

Without collateral fully backing many of the institution's financial portfolios, this kind of risk becomes a major concern for prudent investors. Recognizing moral hazard matters since, under moral hazard, potential asset-poor customers typically face credit rationing—even though they may have worthwhile investment projects. The projects are worthwhile in the sense that they would generate enough income to allow borrowers to in principle pay back loans with interest, and an entrepreneur would profitably undertake the project with their own money if they had the necessary fundsThe potential trouble is that the borrowers cannot find funding for their projects. The same holds for worthy institutions that are rationed out of the capital market due to information problems. Much of the modern theory of financial intermediation has been built upon this particular insight (Diamond 1984; Tirole, 2006) but the application of this theory to understanding the role of public policy and social investors in shaping the emergent financial intermediation landscape remains considerably unexplored.

We argue that commercial investors play important roles, but by virtue of their philanthropic objectives, social investors can help increase leverage, expand scale, and increase total output in a sector in ways that commercial investors by themselves often cannot.

Specifically, we demonstrate cases where philanthropic-giving can in principle raise output, increasing an economy's overall productivity—and lead to GDP increases in addition to poverty reduction. The possibility arises because social investors, by definition, are willing to absorb costs in order to bring gains. This is, of course, their reason for existence. They can do more than shift resources: they can also help increase the total sum of resources available. The question then is:

How can they most powerfully enact such "productivity-enhancing" investments.[1]

Our consideration of effective social investment begins with two ideas. The first consideration is that philanthropic giving can expand the contract space by being willing to subsidize the 'limited liability rents' that often stand in the way of more trades in the commercial loan market. Limited liability rents refer to the returns that must be left with borrowers, or the loan officers that monitor them so as to provide incentives to effort when limited liability constraints restrict the incentives that can be provided by threatening punishments for example by using legal mechanisms to seize collateral. When such rents plus the cost of loan monitoring become too large relative to expected project gains, lenders may walk away from making loans to individuals without collateral wealth, leaving socially valuable projects unfunded. A small subsidy in such contexts can a times be enough to turn a small private loans that might otherwise have been unprofitable now profitable. If enough of those small loans become profitable, a new microfinance institution might emerge where otherwise it might not have.

The second consideration is that effective giving may lead to multiplier effects. It can "crowd in" other resources by catalyzing the investments of others, increasing the recipient's ability to "leverage" its resources. Leverage (i.e. to be able to obtain funds to use in addition to an institution's own equity) is critical as it determines an institution's ability to reach wide scale, and wider scale can lower average fixed costs. Effective social investment aims to expand leverage, holding constant other social concerns.

In this framework, profitability is neither necessary nor sufficient for attracting commercial capital. Merely earning profit will not generate the leverage to take institutions to wide scale. Moreover, productivity can be enhanced by supporting institutions that would not exist if profitability was the sole concern.

The framework has a specific implication for the most effective use of social investment. We describe arguments that push social investment either "high" or "low" when directed to poor communities. Specifically, we show that smart subsidy involves allocations that either go to institutions serving relatively well-off (but still poor) customers, where the potential for leverage is maximized, or to institutions serving the poorest customers. There is less "bang for the buck" in supporting a middle range of institutions that are self-sufficient with their own resources but for which leverage (and thus potential scale) is limited.

2 The Model

The analysis builds upon extensions to a simple model similar to what Jean Tirole (2006) describes as a 'workhorse model' for understanding the modern theory of corporate finance and the role of net-worth in shaping access to credit under moral hazard[2].Our aim is to understand the range of financial contracts and intermediary structures that can and will emerge to serve different segments of a

[1]This language is from Bardhan, Bowles, and Gintis (2000).

[2]Jean Tirole's sweeping 635-page *Theory of Corporate Governance* textbook is built around a workhorse model very similar to this, which he calls the 'simplest model of credit rationing ... to illustrate the role of net-worth (p. 114).' See also Holmstrom and Tirole (1997) and J. H. Conning (1999) for earlier analyses of the underlying model.

heterogeneous population of would-be borrowers, and how this financial landscape might be transformed. To highlight the role of borrower net worth we begin with the extreme assumption that would-be entrepreneurs live in spatially segregated neighborhood according to their initial asset holdings and that localized financial institutions (FIs) or bank branches, managed by knowledgeable local managers, must be established within each neighborhood in order to identify and monitor borrowers. CCreditors to these FIs (which might include depositors) and, later in the paper, informed equity investors may however make investments across neighborhoods.

There will be up to four types of agents: (a) households that become entrepreneurs to run small business firms; (b) one or more local financial institutions in each neighborhood, wholly or partly owned and managed by locally informed equity investors. These financial institutions may lend out of their own equity capital and/or may be able to intermediate funds from (c) private investors (possibly including savings depositors). Finally, social investors may be able to affect the nature and depth of the above relationsips through additional investments, subsidies and guarantees of their own.

Each household is assumed risk neutral and to have access to a production or trading project which requires lump sum investment I to be initiated. If the entrepreneur secures and invests the lump sum and works diligently then the project succeeds and generates cash returns X_i with probability p or it will fail and generate nothing with probability 1-p. If this borrower is non-diligent, for instance because she diverts effort or resources from the project to other uses, then the probability of success is reduced from p to q but the entrepreneur obtains a *private* benefit B. Since the entrepreneurs choice of diligence cannot be verified and private benefits cannot be observed or seized by the lender, the potential for borrower moral hazard arises

We assume potential entrepreneurs have no *liquid* capital of their own, so they must borrow the entire amount I to carry out the project. Would-be borrowers may however have *illiquid* assets A that can serve as collateral guarantee in the event of failure. These are assets the borrower prefers to keep tied up in other projects or investments, or they may be pledgeable future cashflows. What matters is that these resources can be credibly pledged to the lender in the event that the financed project should fail. Although some microfinance lenders do not explicitly require a collateral pledge there is little doubt that most experienced loan officers form an estimate of how much they believe can be credibly recovered from a borrower in the event of low returns on the finance project, and this estimate of A_i will be an essential determinant of the terms of the contracts offered.[3]

An MFI is assumed to begin with initial equity capital K – which we shall often refer to as *local intermediary capital.* The MFI could be either a for-profit financial entity, in which case K would be interpreted as the owners' initial equity investment, or it might be a not-for profit organization which is prohibited from distributing dividends, in which case K might be startup capital, perhaps seeded by a donor.[4] K is capital owned by, or under the control of individuals who know

[3]We assume here that the full value of may be taken from the borrower and costlessly transferred to the lender, but the analysis can be easily extended to cases where some or all of the value of is lost in the transfer.

[4]In later sections we will expand our definition of to encompass certain types of outside equity or

how to operate an MFI in this particular neighborhood and who will be in a good position to monitor the loans of any borrowers that they may create. The point to emphasize is that K is not easily created or simply borrowed from outside of the neighborhood – it is capital that is tied to people with a specific set of skills. Much of the analysis that follows turns on understanding when and how K can be leveraged to attract outside investors to expand the scale of lending in a particular neighborhood.

If K is the MFI's only source of funds (i.e. no outside leverage is possible), then after the MFI has paid for its fixed operating costs F, it will be able to lend out up to $K - F$ worth of funds per period and therefore reach $N_0 = (K - F)/I$ borrowers. Since the MFI always has the option of keeping K invested in a safe investment for gross return γK, and it wants to remain sustainable into the future we will assume that the MFI will only come into business if after considering all cashflows (including possibly outside subsidies) it can expect to earn this rate of return.

Any would-be entrepreneur has the option of not taking a loan and engaging in an alternative activity or occupation where they are assumed to earn a reservation payoff, normalized at zero. A financed project yields either a verifiable return of $X$when it succeeds or it fails to yield nothing. The probability distribution of project returns is determined the borrowers' choice of effort or diligence. If the borrower chooses to be diligent, in the sense that they exert personal effort and use all borrowed funds for the financed activity as stated in the business plan they described to their lender, then the project succeeds with probability p A borrower who is non-diligent lowers the probability of success from p to $q < p$, but captures some private benefits B by avoiding effort and/or by redirecting some borrowed funds to activities whose return may not observed or seized by the lender. B therefore captures the opportunity cost of being diligent.

A loan contract can be described as a contingent borrower reward schedule (s_s, s_f) where s_i indicates the net monetary return to the borrower in state i, which can be success $(i = s)$ or failure $(i = f)$.[5]

We assume that MFIs offer competitive contracts that maximize the expected return to the borrower. This assumption can be motivated either by assuming that each MFI wants to maximize profits but that contestable entry pushes lenders' profits to zero, and/or that MFIs have a social mission that leads them to care about borrower welfare subject only to a sustainability (i.e. break even) condition.

Consider a contract in which a borrower pledges all their available collateral$A$. We focus on this particular contract to identify the 'minimum collateral requirement' that a borrower would have to have in order to be feasibly reached by a lender. The contract will take the form (s_s, s_f) =$(s, -A)$ where $-A$ is the return to the borrower when the project fails (i.e. the project generates zero returns and the borrower loses the collateral pledge). The lender is repaid only $R_f = A$ when the project fails and $R_s = X - s$ when it succeeds. The lender will willingly participate in this contract so long as the borrower is diligent and the expected

quasi-equity investments (such as subordinated debt) from donors or social investors.

[5]With a few additional assumptions the model can be extended to a multiple-outcome multiple-action framework where the optimal contract takes the form of a standard debt contract. This complicates the analysis but does not fundamentally alter the main conclusions (Innes, 1990; J. Conning, 2006)

return under diligence $E[R_i|p] = E[X_i|p] - E[s_i|p]$ exceeds the cost of lending:

$p(X-s)+(1-p)A \geq \gamma I + F/N$

where γ is the gross opportunity cost of funds to the MFI (one plus the interest rate the lender pays) and F/N is the average fixed cost per borrower. For the time-being we take N to be fixed, but later endogenize its value. The line labeled $PC(0)$ in figure 1 indicates combinations of s and A at which this lender's participation constraint is exactly met. A point $(-A, s)$ on this Cartesian diagram represents a potential contract, but only contracts that lie below $PC(0)$ in this contract space satisfy the participation constraint.

The expected return to the *diligent* borrower is:

$$ps + (1-p)(-A) - B$$

$= p(s+A) - A - B$

while the expected return to a *non-diligent* borrower is:

$q(s+A) - A$

It follows that the borrower prefers to be diligent so long as (2) exceeds (3) or

$s \geq -A + \frac{B}{\Delta}$

Where â^†=*p-q*. This constraint restricts incentive-compatible contracts to lie above line $IC(0)$ in figure 1.[6] The range of feasible contracts must therefore lie in the shaded region above $IC(0)$ where the borrower remains diligent and below $PC(0)$ where the lender recovers all costs. Of particular interest is the contract with the *minimum collateral requirement*, or the lowest required pledge A, since this contract will allow reaching the largest group of borrowers. This contract is found at the intersection of $IC(0)$ and $PC(0)$ and is labeled U in the figure.

To see this mathematically note that limited liability sets a floor on the borrower's return in the failure state of $-A$. This and constraint (4) requires that the borrowers' return to success not be set below$s = p/\Delta - A$. A contract that satisfies limited liability and incentive constraints therefore must leave the client with an expected return of at least

$E\left[s_i|p\right] = -A + \frac{\mathrm{pB}}{\Delta}$

Following Laffont and Rey (2003) we refer to expression pB/Δ as a *limited liability rent* (LLR). It is the minimum share of expected project returns that must be left to a client if incentives and limited liability to be maintained. This rent is larger the larger is the opportunity cost of diligence B and the more difficult it is to infer diligence choice from simply observing the project outcome (i.e. the larger is q/p)

Using (5) to substitute for $E\left[s_i|p\right]$ in (1) and then solving for the threshold value of A at which this constraint exactly binds yields an expression for the *minimum collateral requirement* that a client would have to pledge to attract the participation of an unsubsidized lender:

$(0, N) = \frac{\mathrm{pB}}{\Delta} - \left[\mathrm{pX} - \hat{\mathrm{I}}^3\mathrm{I}\right] + \frac{F}{N}$

Where $(0, N)$ refers to the minimum collateral requirement on a zero-monitoring loan when the MFI services N borrowers. Borrowers with assets $A \geq (0, N)$ will be able to obtain loans of size I and earn expected

[6] To keep things interesting we assume parameters such that the lender could not recover lending costs if the entrepreneur chose to be non-diligent. The lender will therefore only be interested in participating if they can be convinced that the contract satisfies this incentive constraint.

return$pX - \gamma I - F/N$, but entrepreneurs with less than this amount will be excluded from the loan market even though their projects may be socially profitable, i.e. $pX - \gamma I - F/N > 0$. For later reference we define $= (0, \infty)$ to be the minimum collateral requirement for a non-monitored loan in a really large MFI (i.e. where fixed costs per borrower are essentially nil).

A higher level of $(0, N)$ implies a larger number of borrowers excluded from the loan market. Expression (6) suggests the available ways for MFIs and other financial innovators to improve loan access and terms: (a) by acting to lower the size of the limited liability rent either by acting to directly reduce the borrower's opportunity cost of diligence B or changing the likelihood ratio (e.g. lending for projects with lower $l = q/p$ or that are 'safer' under diligence compared to non-diligence); (b) by lowering of average fixed costs F/N either by lowering F or expanding scale by increasing the number of borrowers N; or by (c) raising the expected returns from client's projects pX or lowering the MFI's cost of funds γ.

2.1 1.1. Monitored lending

A defining characteristic of most microcredit lending is the heavy use monitoring and other incentive mechanisms to substitute for the use of loan collateral guarantees. One way to interpret monitoring and control activities is that they aim to try to raise p relative to q, for instance via technical assistance that raises the productivity of client efforts under diligence. This approach was followed in Banerjee, Besley, and Guinnane (1994) and Madajewicz (2004). An alternative interpretation is that monitoring of borrowers serves to lower the value of B. This is the approach of Holmstrom and Tirole (1997), Conning (1999) and what we shall follow below.

'Monitoring and control activities' of this last kind aim to reduce the scope for moral hazard by reducing the value of private benefits B that the borrower stands to capture. A straightforward way to capture the role of such active monitoring is to assume that by spending m on monitoring and control activities a MFI can lower the borrowers' benefits of non-diligence linearly from $B(0) = B$to$B(m) = B -$ Î±m, where α is a parameter that captures the degree of monitoring effectiveness. We might think for example of a lender making frequent and unannounced visits to an entrepreneur's home or place of business, or their insistence that the borrower prepare a business plan. These are just a few examples of the costly monitoring activities that can be use to lower a borrower's scope for diverting loan funds and other resources to uses other than the financed activity. Since it doesn't matter whether we define B as a private benefit to non-diligence or a cost to being diligent, we could also think of monitoring as providing helpful technical assistance that also lowers the borrower's cost of raising the probability of project success.

Since monitoring activities are however themselves costly and difficult activities, an MFI's financial backers must put in place the right incentive and governance structures to make sure that their managers and loan officers are properly motivated to carry out this monitoring at optimal levels to pursue the objectives of its primary stakeholders. An MFI's ability to leverage outside funds depends therefore in primarily on how well they can convince outside investors that incentives and monitoring effectiveness will not be damaged once the MFI begins to scale up by using outside funds.

For the moment we shall set aside the question of outside funding and focus first on MFIs that fund their operations using only the internal equity funds K at their disposal. Monitoring by the MFI adds to the total cost of lending (depicted as a shift in the participation constraint from $PC(0)$ to $PC(m_1)$ in figure 2 but it also relaxes the incentive compatibility constraint (depicted as a shift from $IC(0)$ to $IC(m_1)$). With monitoring the earlier expression (6) for the minimum collateral requirement now becomes[7]:

$(m, N) = \frac{p(B-\alpha m)}{\Delta} - \left[\text{pX} - \text{Î³I}\right] + F/N + m$

Under the assumption that on the margin monitoring is a worthwhile activity or that $\alpha/(1-l) > 1$, monitoring will lower the minimum collateral requirement linearly. Figure 2 illustrates how an increase of monitoring from $m = 0$ to $m = m_1$ might reduce the minimum collateral requirement from $= (0)$ to (m_1). Additional monitoring continues to reduce the minimum collateral requirement, but more monitoring increases the cost of funds to the borrower who (from) pays expected borrowing costs

$$\gamma I + F/N + m$$

It should be immediately clear that borrowers prefer less heavily monitored loans because they are less costly. Hence borrowers will chose loans with only as much monitoring as is just sufficient to bring the minimum collateral requirement down to their available asset level$A$. Figure 3 depicts the relationship between monitoring and the minimum collateral requirement . A borrower with available assets $A < A$ would choose a loan with just enough monitoring intensity m_A to bring the collateral requirement down to their available assets, or $(m_A) = A$.

There is a limit to how far monitoring can expand access to asset-poor borrowers. No borrower would voluntarily agree to a loan where the total cost of borrowing exceeds expected project returns. This defines a maximum monitoring level

$= pX - \text{Î³I} - F/N$

beyond which no borrower would willingly participate. To summarize, monitoring can expand loan access to borrowers with assets in the range $A \in [(), (0)]$ that would have been excluded from the non-monitored loan market.

2.2 1.2. Delegated monitoring, leveraged intermediation and scale

Thus far we have said nothing of the organization or governance structure of the MFI, where the MFI obtains its loanable funds from, or what determines the scale of lending (i.e. the number or borrowers$N$).

For each loan of size I, amount I^m will be financed from MFI capital $(K - F)$ and amount $I^u = I - I^m$ (possibly zero) by raising funds from an uninformed outside investor (creditors and/or depositors to the MFI). Raising outside funds offers the opportunity to increase scale from $N_0 = (K-F)/I$ to $N = (K-F)/I^m$ borrowers.

[7]For the time being, we continue to assume that N remains fixed, but we add N as an argument for later use.

Creditors or investors in the MFI might include local saving depositors or national or international investors who make loans to the MFI. We assume that investors have other opportunities for their capital, so they will only be interested in investing if the MFI can credibly guarantee an expected gross rate of return of γ. The central concern of any investor will be whether the management and loan officers of the MFI have the proper incentive to diligently monitor borrowers so as to preserve the quality of the loan portfolio.

There will now be three parties to a loan contract: the borrower, the managers/owners of the MFI, and the outside investor(s). The timing of the contracting problem is as follows: All parties agree to the terms of the contract, the outside lender puts up I^uper borrower, the MFI puts up $I^m = I - I^u$. The client then chooses his unobserved diligence level, understanding and taking into account that, given their stake in his project, the MFI will have incentive to monitor at intensity m per borrower. Project returns are then realized and the generated returns are then divided according to the terms of the contract.

Loan monitoring is itself an activity that is subject to moral hazard. Since the MFI management and staff's actions cannot in general be directly verified by outside investors incentives to monitor must be provided by tying the MFI agents' rewards to observable outcomes on the borrowers' projects. The way outside investors will be convinced that the MFI owners and their staff have sufficient incentives to maintain the quality of the loan portfolio is by demanding that the MFI place enough of their own capital I^m at risk in each borrower's project so as to have incentive to monitor by enough to keep the borrowers diligent.

The MFI's will earn net return $R^m - \gamma I^m - m$ if it monitors and the project succeeds and $-\gamma I^m - m$ if it fails.[8] To have incentives to monitor the MFI must expect to earn more by monitoring at intensity m than by not monitoring at all, where intensity m is the minimum amount required to induce the borrower to be diligent, given the other terms of the contract. The MFI's incentive compatibility constraint is therefore:

$$pR^m - \gamma I^m - m \geq qR^m - \gamma I^m$$

$R^m \geq m/\Delta$

The MFI must also be willing to participate, which is guaranteed if they can expect to earn as much from their efforts and funds inside the relationship as outside (i.e. they expect to earn as much by investing I^m elsewhere as they do by lending:

$pR^m \geq \gamma I^m + m + F/N$

Since the MFI is responsible for organizing the banking activity, including renting buildings and equipment, they must cover F/N to enter and stay in the financial intermediation business.

[8]For the moment we focus on the terms of the contract as if there were only one borrower. In practice the lender will not want to tie the MFI's return to the outcome of a particular project, but rather to the outcomes from the entire portfolio. We will return to this desire for diversification later. For the moment, and to simplify, we follow Holmstrom and Tirole (1997) in assuming that all of the borrowers have projects with stochastic but perfectly correlated outcomes, so all projects succeed or fail at the same time. We will later return to the more general case with diversification, as studied by Conning (2005), Tirole (2005, p158-163) and Diamond (1984).

For the moment we will assume that MFIs are driven by a social concern for the welfare of the borrower (or equivalently that competition amongst potential intermediaries) so there are no economic profits or, equivalently the return on MFI equity is equal toγ. We will later relax this assumption, but for the moment the assumption of zero profits implies that offered loan contract will be on the most favorable terms the borrower, so this participation constraint as well as that of the outside investor will bind in equilibrium. Substituting the intermediary's binding incentive constraint in this participation constraint yields:

$\gamma I^m(m) = \frac{\mathrm{qm}}{\Delta} - \frac{F}{N}$

which defines the MFI's required equity stake in the entrepreneur's project if incentives are to be maintained to monitor at intensity m and lending costs are to be minimized to maximize leverage and outreach.[9] The higher the required amount of monitoring m to keep a borrower diligent, the higher the stake the MFI must hold in the project since $q/$â^†>0. This expression suggests an immediate explanation for why leverage has remained disappointingly low in microfinance. As Ledgerwood (1998, page 41) explains 'most MFIs are not fully leveraged due to their inability to borrow funds on the basis of their performance and debt capacity.' The above expression suggests why: MFIs must monitor their loan portfolios heavily to maintain its value. Monitoring is a replacement for missing collateral. The more heavily the MFI concentrates on a segment of the market without collateralized loans (which could be sold to debt collectors) the more the portfolio value depends on the level of monitoring, and monitoring incentives require the MFI retain a deeper stake and therefore that it be less leveraged. The commonly held myth that the pursuit of higher MFI profits should translate into higher leverage does not take this factor into account.

In a situation where MFIs seek to maximize borrower returns (or in a competitive market) both the MFI intermediary and the uninformed outside lenders (if any can be attracted) earn just enough to cover their costs and earn Î³ in expectation on their investments. The minimum collateral requirement can be found as before, by substituting the borrower's binding incentive constraints the MFI's monitoring incentive constraint and the uninformed lender's binding participation constraint ($pR_u + (1-p)A = \gamma I_u$) into the MFIs participation constraint:

$pR_m \geq \gamma I_m + m + F/NpX - \left[\frac{p(B_0-\alpha m)}{\Delta} - A\right] - \gamma I_u \geq \gamma I_m + m + F/N$

and solving for A the collateral minimizing solution is again as in (7):

$(m, N) = \frac{p(B-\alpha m)}{\Delta} - \left[\mathrm{pX} - \text{Î³I}\right] + F/N + m$

with the important new interpretation that N will now be treated as an endogenous variable. Higher leverage will mean the MFI can reach a larger number of borrowers $N$with any given level of equity $K-F$. This will reduce the average fixed cost of borrowing per borrower F/N which will, in turn, open up space to adjust the terms of the financial contracts and reach yet more borrowers.

For borrowers in asset class A we can now solve for the equilibrium level of borrowers N and the minimum required amount of monitoring per borrower m. These will be given by the solution to the following system of equations:

$(m, N) = A$

$N = \frac{(K-F)}{I^m}$

[9]We later explore the possibility that intermediary capital is scarce and MFIs earn a return on equity above Î³.

where I^m is given by (11). As demonstrated in the appendix this system can be solved to give the following closed-form solution for N and m as a function of a client's initial asset level A and parameters describing the MFI's lending technology and equity base:

$N^* = \frac{\Gamma}{[-A]}$

$m^* = [-A] \cdot \frac{\Delta}{q} \frac{(\gamma K-(1-\gamma)F)}{\Gamma}$

where $= (0, \infty)$ and $\Gamma = \frac{((1-\alpha)p-q)}{q}[\gamma K - (1-\gamma)F] - F$.

Required monitoring falls towards zero as the client's collateral wealth A approaches, the collateral requirement for an non-monitored loan in a big bank (where fixed costs per loan are nil). Leverage, and therefore also the number of borrowers N that can be reached, rise exponentially as A approachesfrom below. This apparently extreme result – that leverage becomes infinite – is an artifact of our implicit assumption that it is costless for the MFI or any third party to collect collateralized debt. In practice local MFIs will have an advantage in collecting compared to an outside investor and there will be moral hazard involved in the collection of such debt if collection is at all costly (**town1979**; Diamond, 1984). It is also true that uninformed lenders (small depositors in particular) will typically not be in a good position to observe the MFI's debt to equity holdings. For all these reasons financial institutions that accept deposits are typically regulated which in effect places an institutional cap on maximum leverage. To capture this idea and to keep the number of borrowers from rising to infinity we will impose an institutional limit on leverage in the form of $N \leq N_{\max}$ This is without loss of generality because the focus of our analysis is on MFIs that will be be unable to reach this high level of leverage anyway.

Figure 4 shows how the equilibrium level of N and m would be determined for MFIs that aim to reach target groups with assets A and A'.

The MFI's balance sheet can represented simply as:

Assets	**Liabilities**
$N \cdot I$ (Loans)	$N \cdot I^u$ (Debt)
	$N \cdot I^m = K - F$ (Owners' Equity)

Of course, when the MFI is unable to lever up and attract outside deposits or other forms of debt the liability side of the balance sheet will consist solely of owners equity (including donations) and the number of loans will simply be $N_0 = (K - F)/I$.

A numeric example will help to illustrate the relationships captured by the model and predictions regarding the reach of different types of lender on this market. Consider a simple case where the parameters are as given in Table 1.

Consider first the earlier case where the MFI lends only out of its available capital base K, without leverage, so $N = N_0$. For these parameters the moral hazard problem would lead lenders to demand $= (0; N_0) = 130$ units of pledgeable income to cover a zero-monitoring loan of $I = 100$. This is the kind of lending carried out by ordinary banks that typically work only with clients with substantial savings or cash flows relative to the size of the borrowed amount. The associated loan would specify fixed repayment amounts of $= (130, 130)$ on a loan of size $I = 100$.

The implied expected cost of funds to the borrower is forty percent: twenty percent to cover the lender's variable cost of funds Î³I and another 10 percent to cover average fixed cost of $F/N_0 = 10$ per loan. The borrower earns an expected return $pX - \gamma I - F/N_0$, which for these parameters is $30 (= 180 - 120 - 30)$.

Consider the possibilities and limits of monitored lending to reach more borrowers. Any borrower with less pledgeable assets A will have to resort to a more expensive monitored loan, or receive no loan at all. Since monitoring raises the cost of borrowing, borrowers will want a loan with only just as much monitoring as is minimally required to bring the collateral requirement down in line with their available assets A, or the minimum m to set $(m, N_0) = A$ using (7). For instance a borrower with $A = 60$ would require a loan with minimum monitoring intensity $m = 35$, assuming an MFI with no leverage, or $N = N_0$.The borrowers total cost of borrowing I will then be $\gamma I + F/N_0 + m = 120 + 20 + 35 = 175$, for an implied interest rate of 75 percent.[10]

Hence, although monitoring expands access, these asset-poor borrowers pay much higher financing costs to cover the MFI's higher costs of monitoring. For the parameters above, the maximal feasible level of monitoring is $= 50$ and the lowest the collateral requirement can be pushed is down to $() = 30$. Monitoring carried out by the MFI with no leverage can therefore expand access to borrowers with assets in the range $A \in [30, 130]$ who would otherwise not have been served. Any borrower with A less than () will prefer to forgo a loan and remain in the low-return occupation. Even though they have access to a socially profitable investment (and one that they would carry out if they had had their own funds) they are unable to secure financing to profitably carry out the project due to the the nature of the information asymmetry.

In the example thus far we have assumed that the MFI uses only its own funds. It has $K = 11000$ of capital, $F = 1000$ of annual fixed costs, and it leverages no outside funds. It was therefore able to lend out only 10,000 through $N_0 = 100$ loans at I^m =100 dollars per loan). This MFI's balance sheet would simply look as follows:

Assets		**Liabilities**
10,000 ($=N_0 I^m$)	Loans	10,000 ($= K - F$) Owners Equity

Consider next the case of an MFI that attempts to leverage outsider's funds to increase outreach and lower average fixed costs. Notice immediately from (11) that if there is no monitoring required there is no minimum required stake I^m and leverage $(I - I^m)/I^m$ could in principle be infinite. In that case the MFI would be a perfect and costless intermediary who could always repay its investors, because it has fully collateralized loans and zero costs of intermediation. This would be the financial equivalent of a perpetual motion machine. Since, as previously discussed, even fully collateralized loans require some monitoring and collection is itself a costly activity, in reality I^m would never be zero, if for no other reason than that regulators would intervene to limit leverage via capital adequacy requirements.

[10]While this borrower pays out only 60 out of collateral when their project fails (compared to 130 for the earlier borrower who could post collateral to receive a loan with zero monitoring), they take home only 23.33 when the project succeeds (compared to 70). This leaves this borrower with an expected profit of only 15 (compared to 30).

We will assume that regulators require MFIs to have a least ten cents of their own capital at risk for every dollar loaned (i.e. $I^m \geq 0.1I$) , or that they can borrow no more than ten times their equity base.

For the above parameterized example this means that $N_{\max}$ With this many borrowers the minimum collateral pledge on a non-monitored loan drops to $=$ $(0; N_{\max}$ (from 130 when$N = N_0 = 100$) on account of the fact that the average fixed cost per borrower has been reduced from $F/N_0 = 10$ to $F/N_{\max}$.

A collateral requirement of 121 on a loan of size 100 still leaves a lot of excluded borrowers, so let's turn to how increased leverage expands the possibilities of using monitored loans. The optimal monitoring intensity m and maximum feasible scale N is determined by equations (16) and (15) and will depend on the level of A of the target borrower group. Starting from the zero-monitoring collateral requirement , we can lower the collateral requirement to reach successively poorer borrowers with assets $A <$ by increasing monitoring intensity m. As monitoring intensity rises however, the minimum investment stake of the MFI I^m must also rise (given by (11)). This means that MFIs that specialize in reaching neighborhoods with lower A achieve lower leverage I/I^m and therefore also that they reach a lower number of borrowers N with a given amount of startup capital (expression ((14).

Figure 4 describes the financial landscape that emerges in this model. We assume that there are many borrowers at each asset level in each neighborhood and a different MFI (with initial capital K and fixed costs F) specialized in lending to borrowers in each of these neighborhoods. In the emergent landscape neighborhoods will be partitioned into four groups, depending on the initial asset level A of the household in each neighborhood. The figures show the relationships between pledgeable assets A of the MFI target group borrowers and A) the number of borrowers that the MFI can reach with its available capital K, B) net returns per borrower, and C) total returns (net returns per borrower times number of borrowers). Neighborhoods will be characterized as falling into one of the following groups:

Group I – Excluded Borrowers $(A < A_{\mathrm{LO}})$

Entrepreneurs with assets A below threshold level $A_{\mathrm{LO}} = (, N_0)$ cannot obtain financing at any profitable rate. As a rule MFIs can only reach asset-poor borrowers via increased monitoring, but these borrowers are so poor that the monitoring required would be prohibitively expensive. The threshold level of monitoring was earlier described in (8), as $= pX - \gamma I - F/N_0$. In the absence of any subsidies MFIs that try to reach borrowers with $A < A_{\mathrm{LO}}$ would lose money and eventually go out of business.

Group II – Monitored loans from non-leveraged MFIs $(A \in [A_{\mathrm{LO}}, A_{\mathrm{LEV}}])$

MFIs that service borrowers with assets in this range use monitored lending but cannot attract any outside investors because the minimum required stake I^m in (11) that is required to attract an outside investor exceeds the total value of the loan I. These MFIs can reach asset-poor borrowers using their own capital, or $I^m = I$ but these MFIs can reach only as many borrowers $N_0 = (K - F)/I$ as their capital permits. The net expected return to a borrower is rising with asset levels over this range (since borrowers with more collateral require less expensive monitoring), as depicted in the top panel of the figure.

These MFIs cannot sell any part of their portfolio of monitored loans without, in the eyes of outside investors, diluting their incentives to monitor the quality of that loan portfolio diligently. MFIs trying to reach borrowers with assets in this range cannot attract outside investors not because they are not profitable (in fact they are earning normal profits, as are all other MFIs). The commonly heard claim that leverage would be raised if only the MFIs raise profits further by raising the borrowers' required repayments can be seen to be flatly untrue here. In fact, demanding a higher repayment of borrowers would only exacerbate rather than improve the situation, by aggravating the moral hazard problem for borrowers.

Group III – Monitored Loans from leveraged MFIs ($A \in [A_{\text{LEV}}, A_{\text{HI}}]$)

MFIs that service borrowers in this range are able to leverage their capital, and therefore reach a larger number of borrowers. This is because the required minimum equity stake I^m is less than the full amount of the loan I. These MFIs can therefore reach $N = (K - F)/I^m > N_0$ borrowers, and this number of borrowers increases exponentially with higher the level of A (up to $= (0; N_{\max}$ as depicted in the middle panel of the figure. The borrower's net return rises more quickly with A than for borrowers served by non-leveraged MFIs (top panel of the figure) because the costs of borrowing fall more rapidly with rising A because these MFIs can take advantage of increasing scale (which lowers average fixed costs). The cutoff level $A_{\text{LEV}} = (m_{\text{LEV}}; N_0)$ where m_{LEV} is given by the m at which $\gamma I = \frac{\text{pm}}{(p-q)} - m$.

Group IV – Non-Monitored Loans from highly leveraged MFIs ($A \geq A_{\text{HI}}$)

The lower panel of figure 4 shows total borrower gains from the presence of one MFI with $K = 1000$ in each neighborhood. More value is created in the less asset-poor communities because MFIs that operate there have lower costs due to less monitoring and larger scale due to the possibility of outside leverage. This is what the financial structure of many markets looks like: the private financial system serves lots of people in those segments of the market where borrowers have lots of assets or plegeable income (e.g. stable wages that can be garnished) but fewer or no borrowers as we move 'downstream' to segments of the market where borrowers have less pledgeable assets, because they are asset poor or self-employed. Of course we might not expect there to be neccessarily the same amount of 'intermediary capital' K available in each segment of the market, but we will start with that assumption in order to explore how additional contributions of capital or subsidy will have an impact on different segments of the market.

2.3 1.3. 'Smart' subsidies

The graphs point to a major challenge confronting microfinance institutions and their financial backers. Donors may be inclined to target capital toward MFIs that focus upon asset-poor clients and the poor because donors typically care more about the social welfare of poor individuals, but the figure and the model suggests that each extra dollar of capital transferred to higher-end MFIs can 'crowd in' more private sector resources than the same dollar targeted to an MFI trying to reach the poor.

Before examining how a philanthropist or donor might choose when faced with these tradeoffs and whether it might choose to direct its resources via subsidies, grants or loans, it will be worthwhile to briefly pause to clarify a few terms. There is a great deal of debate and sometimes confusion in the policy literature over the role of subsidies. History, theory and evidence indicate clearly that subsidies often do more harm than good. A badly designed subsidy leads to a soft budget constraint problem which creates dependence and weakens rather than strengthens MFI incentives to look after loan portfolio quality. But in models such as this where problems of asymmetric information and limited liability create market failures, a carefully chosen subsidy can create markets where they did not previously exist and crowd in resources in ways that may raise rather than lower total surplus.

There may therefore be 'smart' ways to deliver subsidies and a primary aim of this framework is to help identify where such 'smart subsidies' might be possible. Consider for example how optimal contracts could be adjusted if an MFI stepped in to subsidize fixed costs. So long as the subsidy is delivered in a way that leads parties to recontract to maintain incentives then outside investment, MFI outreach and borrower welfare could well be enhanced. To fix ideas, consider the case of a donor that had \$500 funds to partially subsidize the fixed costs of just one MFI, reducing these from $F = 1000$ to $F^{'} = 500$. Which MFI should the philanthropist target for 'maximum impact?'

In debating this issue it is important to draw a distinction between levels and changes. The panels in figure 4 indicate levels and clearly show that (in the absence of subsidies) an MFI with \$1000 of capital K focusing on asset-rich borrowers can reach more borrowers (N_A) and generate more total net surplus ($G_A = pX - \gamma I - m_A - F/N_A$) than another MFI with the same \$1000 of capital K focusing on asset-poor borrowers.

This might seem to suggest that if a philanthropist had another \$500 to allocate, they might want to deliver the subsidy to an MFI that targets the relatively more affluent, since each dollar of subsidy can be used to 'leverage' outside funds. But this ignores the fact that the \$500 can also 'leverage' capital in a different way. Suppose that in a poor community there are local investors with local intermediary capital K who, given existing fixed costs, information problems and the average net worth of borrowers in this neighborhood, have not found it profitable to pool these resources to form an MFI, but that with a \$500 subsidy to fixed costs these investors could earn enough to want to enter business. In this case the \$500 subsidy would have 'leveraged' in local capital that brought into existence an MFI that can serve N_0 borrowers, where no borrowers were previously attended.

The panels in Figures 5 and 6 illustrate the possibilities using our parameterized example. The panels in figure 5 reproduce those in figure 4 (lines in red), but now also draw in blue dotted lines that represent outcomes from an alternative scenario where each of the MFIs (or potential MFIs arranged according to the asset levels A of the borrowers they attend) would respond to a subsidy that lowered its fixed costs from 1000 to 500. In analyzing the effects of these interventions it will be useful to classify MFIs and the neighborhoods A in which they reside into 'impact groups' according to the type of transformation they undergo as a consequence of an intervention:

Impact Group	Definition	Description
A	Group I $\rightarrow$ Group II	New MFIs emerge where none previously existed.
B	Group II $\rightarrow$ Group II	Supports existing MFIs, no new leverage.
C	Group II $\rightarrow$ Group III	Transforms non $-$ leveraged into leveraged MFI.
D	Group III $\rightarrow$ Group III or IV	Increased leverage in already leveraged MFI.

As can be seen a subsidy delivered to every neighborhood would shifts A_{LO} to the left, with the effect of creating impact group A by bringing into existence new MFIs to serve asset-poor neighborhoods where none existed before. It also lowers A_{LEV}, allowing previously non-leveraged MFIs in (Impact Group C) to now access the private debt markets for the first time. Through subsidies (in the form or outright grants, subsidized loans, loan guarantees, etc) donors can bring new MFIs into existence, they can help transform existing non-leveraged MFIs into leveraged intermediaries that are now better able to raise funds from private markets. Finally they can help increase leverage in MFIs that can already raise private debt.

The question remains how should donors decide to target their scarce funds across differnt MFIs and Impact groups? The incremental impact of these subsidies can be seen in figures 5 and 6. Figure five shows variable levels before and after the intervention, while figure six looks at this difference, and decomposes it into component parts. The top panel of figure 6 shows ΔG_A, the marginal impact of the subsidy on the net returns of a single borrower in each neighborhood, where $G_A = pX - \gamma I - m_A - F/N_A$ and $\Delta G_A = G^{'}_A - G_A$. The next panel shows the number of new clients created in each neighborhood $(N^{'}_A - N_A)$ and the third panel shows the total gain from the intervention $G^{'}_A \cdot N^{'}_A - G_A \cdot N_A$, in each segment. This can in turn be decomposed into gains to existing clients, given by the number of existing borrowers in that segment N_A times the change in net return $(G^{'}_A - G_A)$ per borrower, and gains to new clients, given by the change in number of clients $(N^{'}_A - N_A)$ times the net return to each of these new clients $G^{'}_A$ or:

$$G^{'}_A \cdot N^{'}_A - G_A \cdot N_A = N_A \cdot (G^{'}_A - G_A) + G^{'}_A \cdot (N^{'}_A - N_A)$$

The panels shows that the impact of a subsidy on a representative borrower in each MFI is not at all uniform across neighborhoods and impact groups. Even though the new MFIs created in the lower Group A segment of the market are not able to leverage funds from outside sources, and they could not survive without the subsidy, the donor's funds have 'crowded-in' *local* intermediary capital, and by doing so have helped borrowers with socially profitable projects to become entrepreneurs. As can be seen many more new clients were created by targeting this subsidy to Impact Group A neighborhoods than to Impact Group B and C and many (but not all) Group D neighborhoods. If a donor's main objective is to brag about the number of new clients its intervention has helped to create then donors should focus their efforts on MFIs in poor neighborhoods. The number of new clients in neighborhoods in Group B hardly increases at all, because these MFIs do not achieve leveraged. The subsidy allows the MFI to increase the pool of funds available for lending by 500, allowing five new borrowers to be reached with a $I = 100$ loan each. The impact is larger in all other cases because the intervention leverages in additional sources of financing to expand scale.

Any proper analysis of impacts should however not just count new borrowers but also consider the impact of the intervention on individual borrowers' incomes. As the bottom panel of figure six indicates the per borrower impact group A rises rapidly with the borrowers' initial asset holdings$A$. This is because the poorest amongst this group of all new borrowers will have most of their expected project returns eaten up by the very high monitoring costs needed to bring them into the market. The return left to the borrower therefore rises as monitoring costs drop linearly with higher A.

Focusing on the next impact group B (i.e. borrowers served by MFIs that began as un-leveraged Group II neighborhoods and remain unleveraged after the intervention) we find by looking at the lower panel that most of the gains come from extending a small gain to a an existing group of borrowers, rather than from new borrowers. That is because these are borrowers who already had access to capital, and since no leverage is achieved, scale is increased only very modestly. Each of these borrowers is benefited by the fact that the average fixed cost per loan is reduced by $500/N_0(= 5)$ per borrower plus an additional amount of savings from the need for a bit less monitoring at each level of A. In Impact Group C on the other hand the number of borrowers starts to rise more rapidly with A because the multiplicative impact of leverage. As drawn, and for these parameters, Group C is not a very large number of borrowers.

The analysis of group D is interesting. This impact group consists of MFI that already achieved leverage, but with the aid of a subsidy are now able leverage further. The multiplier effect of the subsidy via leverage shows up primarily as an exponentially rising increase in the number of new clients. The impact per client is however declining with A. This is because were already in very highly leveraged institutions where average fixed costs per borrower were low and borrowers were already not very heavily monitored. Increased leverage therefore shaves relatively little off the already low cost of reaching these borrowers. As drawn however this dwindling impact is being multiplied by a fast rising population of new borrowers to yield an overall rising total impact of the subsidy as one targets more affluent neighborhoods.

The graph on the lower panel of figure 6 also draws in a horizontal line at 500. If we were being strict utilitarians and measuring benefits as unweighted dollar value of total gains by neighborhood, we would conclude that expected benefits to society of the intervention do not compensate for the 500 subsidy cost for the very poorest borrowers. Of course this will not typically be the welfare criterion a donor might choose to pick since a donor is more likely to place higher weight on gains created in poorer households.

Note that the free market outcome in the absence of philanthropists and their subsidies (including subsidized investments) is constrained efficient. That is, given the initial distribution of assets, structure of the economy including information asymmetries, and the assumption of voluntary exchange, there are no pareto-improving redistributions. Entrepreneurs cannot commit to repaying others in society for loans or transfers that can raise total output, because they cannot commit to making such repayments without undermining their own incentives to maintain the required level of effort. When however a new Bill Gates emerges to become a philanthropist or a donor however, total output can be increased because he does not demand full repayment (and indeed the transfer would be Pareto-improving since the subsidized borrower gains and Bill Gates is made no worse off since the contribution was voluntary).

2.4 1.4. Profits, ownership and incentives

2.5 (this section very rough)

So far we have assumed the MFI behaves so as to maximize outreach and with that lower costs and increase benefits to pass on all the gains in surplus to the average borrower. This would be consistent with the behavior of a non-profit that cannot distribute profits and whose incentives are aligned to maximize borrower welfare because of the 'social mission' orientation of its principal shareholders. The MFI is able to sustain itself over time but the rate of return on equity is Î$^{\mathbf{3}}$, the same rate return as investments in other areas of the economy.

An intermediary with a profit motive will seek to earn an expected rate of return on equity in excess of Î$^{\mathbf{3}}$.

$p\frac{m}{\Delta} = \beta I_m + m + \frac{F}{N}$

We can interpret Î$^{\mathbf{2}}$ as the expected return on equity:

The ROE is Î$^{\mathbf{3}}$ when intermediary's IC constraint binds. If however intermediary capital is scarce, they will be able to earn a rent Î$^{\mathbf{2}}$>Î$^{\mathbf{3}}$. This will be associated with a lower I_m (i.e. higher leverage). Appendix 2 explores this scenario in further detail...

2.6 1.5. Extensions

As in all models we have made simplifying assumptions to highlight important mechanisms, but these simplifications may at times bias the conclusions. A case in point is the assumption we have made that within each neighborhood resides an unlimited pool of identical entrepreneurs each with a high social return project (i.e. one that generates 200 for success with probability 0.9 when the borrower is diligent). This assumption tilted the conclusion of our analysis to favor subsidies and social investment toward higher-leveraged MFIs because under these assumptions these MFIs could pull in a large number of new borrowers. A more realistic assumption is that the distribution of returns across borrowers is heterogeneous, and that the marginal quality of the entrepreneur's project declines as the MFI expands scale in a particular neighborhood.

Under this new set of assumptions the social investors' tradeoffs are likely to be rather different. Suppose for example that instead of all borrowers having a project with a certain return of $X = 200$ under 'success' this return where itself a random number drawn from a distribution. This distribution is assumed to be identical across neighborhoods and lenders in each neighborhood can observe X prior to giving a loan. It is easy to see that under the private market solution with no subsidies, all else equal, clients with lower X will receive more heavily monitored loans, and lenders will extend scale by moving from higher X borrowers to lower X borrowers to maximize total surplus per neighborhood. It should also be clear therefore that in a higher Aneighborhood the last borrower to be reached has a lower X compared to a poorer neighborhood where there is less leverage. Under these circumstances the tradeoffs faced by a donor considering where to place a subsidy are now different. They are now more likely to tilt their decision toward poorer neighborhoods because that is where they will find borrowers with higher expected social marginal returns.

3 Conclusion

The act of making an investment in a microfinance institution typically changes the incentives for the institution's owners and managers. The investors now take on risk that had previously been borne more heavily by the microfinance institutions. The institutions and their managers now face fewer (or different) consequences of their actions. This shift in who bears the risk for successes and failures can seriously diminish incentives for diligence on the part of institutions and their staff—such that a past history of successes may be a poor predictor of future performance.

Drawing on the discussion of moral hazard above, we created a landscape of outcomes. In a free market system, without social investors, we expect that the landscape of finance for the poor would be divided into four broad situations.

1. *The highly-leveraged.* The first involves institutions lending to better-off borrowers that have collateral—and where courts function effectively to seize assets. Here, microfinance should be maximally leveraged since outside investors can invest with limited concern about the diligence of the institution's staff. Loans do not have to be monitored in the same way that un-collateralized loans do. In the worst case, if customers default on their obligations, collateral will be there as a back-up. More re-assuringly, the worst case is unlikely to materialize since the fear of losing collateral should give customers adequate incentives to repay loans. Investors will still worry about currency risk, political risk, or macro-economic/environmental risk, but investors can feel confident that the fundamental investment should be unaffected by the institution's management structure, ownership rights, or degree of leverage.

2. *The moderately-leveraged.* A second group of institutions serves borrowers with less collateral than the first group. The institution has to monitor and take other special efforts in order to obtain ensure loan repayment. Given moral hazard, the more monitoring that is required, the more the institution has to have incentives to monitor. Here, the institution's owners have to be exposed to some of the risk associated with the failure to monitor sufficiently. If they didn't, potential investors would hesitate to invest. The microfinance institution can thus achieve some leverage, but not as much as would be possible when all loans are fully collateralized.

3. *The non-leveraged.* A third group of institutions serves borrowers with even less collateral. Incentives are such that no external leverage is possible at all. The institution is sustainable at its existing scale. The institution mobilizes resources from its own customers and proceeds on that basis—but commercial investors fail to invest.

4. *The un-banked.* The fourth group of institutions are really not institutions at all. They would fail to exist in a free market system. Monitoring costs are so high that institutions would lose money servicing customers. The customers may have worthy projects, but they lack basic assets and are perceived to be so risky that they remain un-banked. The potential customers would only be served if subsidies emerged to launch the institutions.

We argue that there are two kinds of leverage that matter, and that this recognition should push social investors either to focus on the lower-end of markets that serve the poor or, for quite different reasons, to focus on the higher-end. The first kind of leverage involves bringing new pro-poor institutions into being but does not draw in commercial capital. These are the latent institutions identified in the fourth category above. Once in existence, the institutions can tap the capital of their customers, and perhaps of others in the region, but they are unlikely to be able to tap outside capital. By helping to bring these institutions into existence, social investors can help put the local capital to use and, in that way, multiply the power of their gift.

The second kind of leverage involves social investments that can "crowd in" commercial capital to a maximal extent and the focus here is the second category above. The first category of institutions, those that are already highly leveraged in free markets, need little boost to reach scale.

There is a middle ground in which little leverage is likely, captured by the third category above, where institutions already exist in the free market but they are not well-positioned to access commercial capital. Social investments in the middle ground will have far less of a multiplier effect than investments directed either up-market or down-market. These institutions serve their customers well, and they manage to tap their customers' deposits to maximize scale. But the institutions are poorly-positioned to expand rapidly because of the inherent riskiness of their portfolios and costs of their business, and, if rapid expansion is the goal, social investors should look elsewhere.

Two points should be emphasized. First, social investment allocations should also depend on the weights that investors attach to aiding the various populations under consideration. Those weights may temper the results here but will not affect the basic tradeoffs.

Second, we have assumed, without stressing it, that social investors are not insiders in the institutions receiving funds. Investors have no special knowledge and no special clout. In the language of corporate finance, they constitute "uninformed" capital. The investors may be Silicon Valley entrepreneurs, European religious groups, Japanese aid agencies—each captured by the desire to maximize the social impact of their resources, but retaining some distance from the ultimate beneficiaries.

But imagine instead that investors have good information about the diligence and management of the institution receiving the investment, and that they have clout to affect the institutions' direction. If investors are uninformed, they will have little means to instill good incentives or mitigate moral hazard. If they are informed, though, matters can be improved. In a world of limited information, uninformed investors dominate the market (and they are vital in reaching scale), but informed investors can have particular influence. The power of informed social investors comes when they are able to combine capital with knowledge, and in that way are able to attract "uninformed" investors. Incorporating this element into the framework will not change the basic conclusions above, but it could open up new practical strategies.

Little analytical has been written on social investment, or for that matter on the economic incentives and constraints that guide philanthropy. We have looked instead, partly out of necessity, to a very different literature that shares important challenges and questions: the economics of corporate finance. In building analytical links and developing a simple framework for discussing new ideas, we hope to spur others to take the analytical challenges seriously.

4
Appendix 1

Closed form solutions for N and n are obtained from the system of equations

$(m, N) = A$

$N = \frac{(K-F)}{I^m}$

Substitute $I_m = \frac{1}{\gamma}\left[\frac{\mathrm{qm}}{\Delta} - \frac{F}{N}\right]$from into and the result into to solve for m:

$$N = \frac{(K-F)}{\frac{1}{\gamma}\left[\frac{\mathrm{qm}}{\Delta} - \frac{F}{N}\right]} N = \gamma\Delta N \frac{(K-F)}{[\mathrm{Nqm} - \Delta F]} m = \frac{1}{N}\frac{\Delta}{q}\left[\gamma K + (1-\gamma)F\right]$$

Substituting this back into to and re-arranging yields:

$$(0) + \frac{F}{N} + m\left[1 - \frac{\text{Î±p}}{\Delta}\right] = A(0) + \frac{F}{N} + \left[\frac{1}{N}\frac{\Delta}{q}\left[\gamma K + (1-\gamma)F\right]\right]\left[1 - \frac{\text{Î±p}}{\Delta}\right] = AF + \left[\frac{\Delta}{q}\left[\gamma K + (1-\gamma)F\right.\right.$$

$$\boxed{N = \frac{\Gamma}{[(0) - A]}}$$

Where $\Gamma = \frac{((1-\alpha)p-q)}{q}[\gamma K - (1-\gamma)F] - F$.

The closed form expressions for m is then given by:

$$\boxed{m = [(0) - A] \cdot \frac{\Delta}{q}\frac{[\beta K + (1-\beta)F]}{\Gamma}}$$

Hence in equilibrium m rises proportional to the gap between $A(0)$ and A – poorer borrowers need more monitoring in equilibrium. The number of borrowers reached by an MFI varies inversely with the gap between $A(0)$ and A

APPENDIX 2

('Selfish equity' extension)

2.A Selfish equity, no heterogeneity in each neighborhood

In the current setup the rate of return on equity Î$^{\mathbf{2}}$ was assumed to equilibrate to Î$^{\mathbf{3}}$, the same rate of return outside uninformed investors expect on their investments. This assumption could be justified by assuming that MFIs are run by social investors willing to accept such return. This appendix examines the alternative scenario where it is assume that the overall amount of 'informed equity' (i.e. capital held by individuals capable of managing a financial institution) in the economy is limited, but that this informed equity can move across 'neighborhoods' in search of the highest return.

In order to isolate effects we make the additional simplifying assumption that each neighborhood has the same finite number $\overline{N}_j$of would-be entrepreneurs with access to the same lump sum investment project which gives gross return X when it succeeds and zero otherwise. In a world where investors are purely profit driven, the first equity investors would at first want to direct their scarce resources and talent to serving the wealthiest neighborhoods for the simple reason that there is more project surplus, and therefore economic rents to be made, where resources are not being used up in costly monitoring. Competition for wealthiest clients should drive down the rate of return in the wealthiest neighborhoods and lead to a spillover of investments into less wealthy neighborhoods such that in equilibrium the market rate of return on equity should be driven to an economy-wide equilibrium level Î**2** which is the same in all neighborhoods where clients are served.

To see how this equilibrium is reached, we first we allow the expected rate of return in a given neighborhood to be set at arbitrary level Î**2** and later solve for an equilibrium where Î**2** that clears the market for K.

When there are fixed costs to setting up an institution one should really have to be worried about strategic behavior and timing. In the math below we have included fixed costs per borrower in most expressions to allow for the possibility of analyzing this at some later date, but many of the arguments really implicitly assume F=0.

The main result of interest is that the pursuit of higher profits changes the conditions of access and will in general lead to 'over-monitoring' in the market. To see this note the rate of return on equity is:

Or put only slightly differently, the expected earnings to the equity investor/monitor from an incentive-compatible contract must exactly equal the opportunity cost of those funds I_m in other equity investments:

$\beta I_m = \left[\frac{q}{\Delta}m - \frac{F}{N}\right]$

Since the rate of return Î**2** demanded by for-profit equity investors will in general exceed the rate of return Î**3** that equity owners obtained in the earlier analyzed 'most-generous social investor' case, borrowers (who were contributing as much collateral as they had in that earlier scenario) must now be more heavily monitored if incentives are to be kept in place for them to credibly commit to these higher levels of expected repayment. To see this formally note that the contract must now divide expected project claims pX between sufficient expected payments for the uninformed investor (who puts up$I - I_m$) to participate, sufficient expected returns for the borrower to be diligent, and expected payments $pR_m = \frac{p}{\Delta}m$ to the monitor/equity owner:

$pX = \gamma(I - I_m) + \left[\frac{pB(m)}{\Delta} - A\right] + \left[\frac{p}{\Delta}m\right]$

Re-arranging to solve for A as before we get:

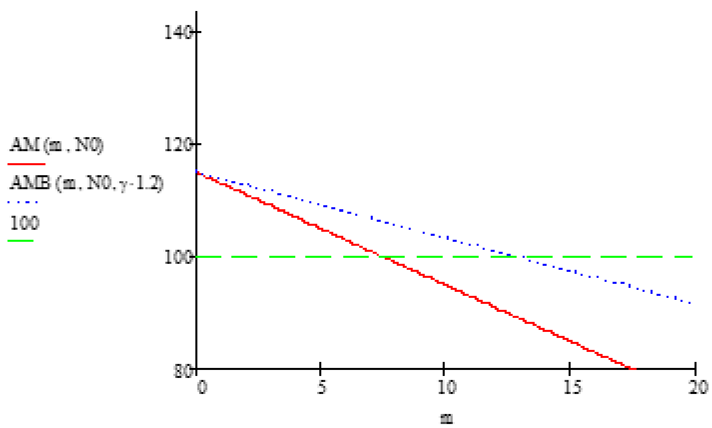
where(m, γ) was the minimum collateral requirement in the 'generous social investors' case where the rate of return on equity was assumed to be Î**2**=Î**3** (i.e. equity investors accepted the same rate of return as uninformed investors).

Substituting for I_m from above allows us to further re-arrange:

Which is all a lengthy way to say $(m, \gamma) = (m, \gamma) + (\beta - \gamma)I_m$

**** COMMENT ***

The following figure illustrates how the monitoring intensity must rise. The lower line is the minimum collateral requirement for the social investor case (i.e. Î^{2}=Î^{3}=1.1), while the top line is the minimum collateral requirement is 20% higher, or Î^{3}=1.32). A borrower with say $A_j = 100$would obtain a loan with (approximately) 7.5 units of required monitoring in the first place, while the same borrower would require 12.9 units of monitoring



*** END COMMENT ***

'Closing' the model

Suppose that there are $j = 1...J$ neighborhoods with $\overline{N}_j = N$ potential entrepreneur/borrowers in each neighborhood, who differ only in terms of their collateralizable wealth A_{j}.

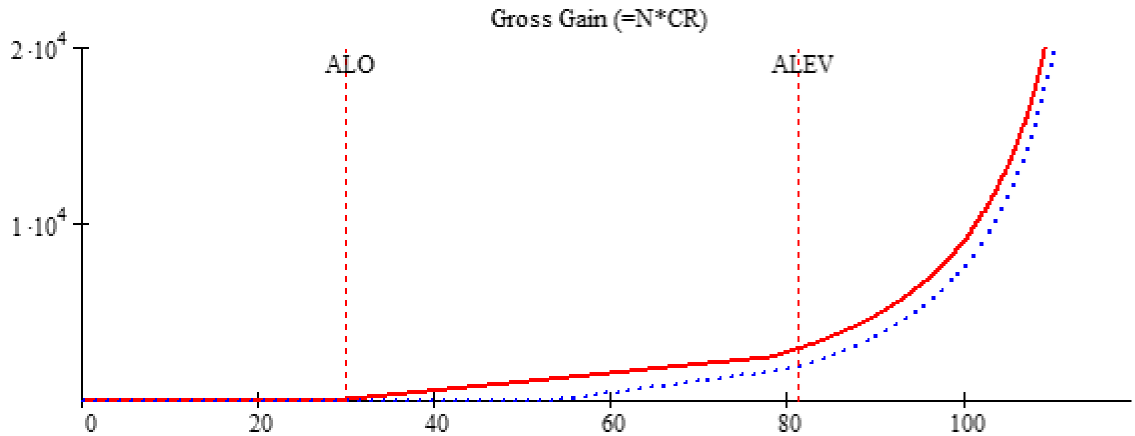
Equity investor funds will flow to the wealthiest neighborhoods until all borrowers are served. Competition will however drive down the rate of return in this neighborhood making it attractive for funds to flow to the next richest neighborhood, and so on.

The logic runs loosely like this. Suppose there were just two neighborhoods, one rich, one poor. Suppose K is in very scarce supply. It's obvious that these equity investors will setup a bank in the richer neighborhood since that is where they can leverage more outside investment, and hence earn a higher return on equity. These profit-seeking investor would like to be as highly leveraged as possible, but competition amongst them to attract rich borrowers will force them to be less highly leveraged

Any remaining funds will flow to the next richest neighborhood until all borrowers are served, and so on, flowing to the next richest neighborhood each time. In the last neighborhood served

Conceptually we can solve for demand for K in each neighborhood as follows. Take neighborhood A_j . From

How does shape change if Î^{2}>Î^{3}: If the MFI demands a higher return on equity Î^{2} (perhaps because monitoring equity capital is scarce in the economy – we can endogenize this rate of return by specifying a fixed quantity of monitoring equity in the economy) the higher share of project surplus devoted to equity payments shifts the thresholds and reduces access.



Access is reduced (not surprisingly since the MFI is taking more surplus off the table). Note that there is also now a higher level of monitoring at each level of A. This is because if the MFI is to take more money off the table, the borrowers must have incentives to be able to commit to deliver it. Or put differently, with less money for them to take home when things succeed their incentive to be diligent would be diluted, unless monitoring can be stepped up.

THERE ARE TWO EXPRESSIONS IN TWO UNKNOWS (m, N):

$$N = \frac{K-F}{I_m}$$

Substituting in $I_m = \frac{1}{\beta}\left[\frac{p \cdot m}{\Delta} - m - \frac{F}{N}\right]$ these can be rewritten in terms of just N and m:

Note that if $\hat{\text{I}}^2 = \hat{\text{I}}^3$ the expression reduces to:

$m = \left[\frac{(p-q)}{(\alpha-1)p+q}\right]\left[[_0 - A] + \frac{F}{N}\right]$

AND

$$\boxed{m = \left(\frac{\beta K + (1-\beta)F}{N}\right)\left(\frac{p-q}{q}\right)}$$

Solving for N

$$\beta K + (1-\beta)F = N\left[\frac{\hat{\text{I}}^2\text{q}}{(\alpha-1)\beta p+\gamma q}\right]\left[[_0 - A] + \frac{\gamma}{\beta}\frac{F}{N}\right][\beta K + (1-\beta)F]\left[\frac{(\alpha-1)\beta p+\gamma q}{\hat{\text{I}}^2\text{q}}\right] = N\,[_0 - A] +$$

From here I should be easy to show that, all else equal, an increase $\hat{\text{I}}^2$ lowers equilibrium N and raise equilibrium m.

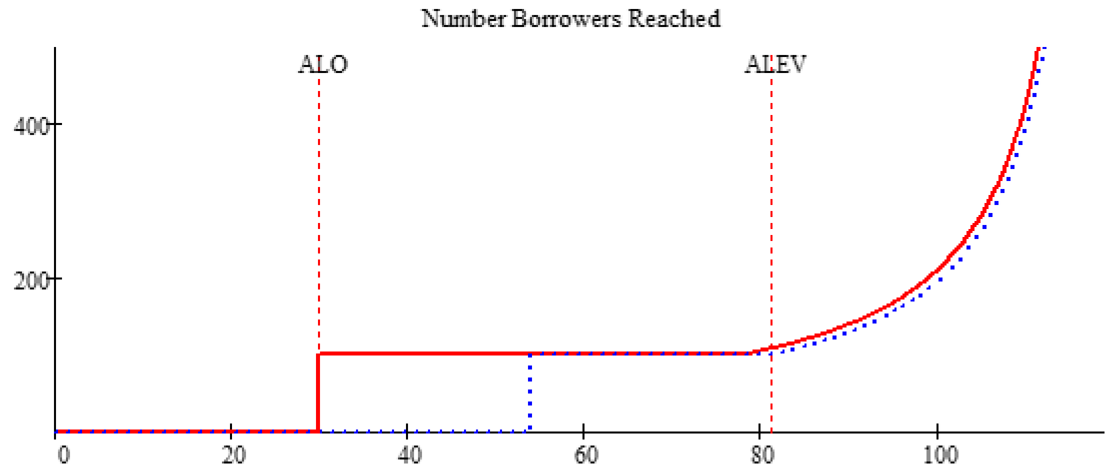
Borrower's participation constraint

$\text{pX} - \hat{\text{I}}^3\text{I} - \left[\frac{\text{pm}}{\Delta} - \gamma I_m\right] \geq 0$

Which can b rewritten

$$\text{pX} - \hat{\text{I}}^3\text{I} - \left[\frac{\text{pm}}{\Delta} - \gamma I_m\right] \geq 0\text{pX} - \hat{\text{I}}^3\text{I} - \frac{\text{pm}}{\Delta} + \frac{\gamma}{\beta}\left[\frac{\text{pm}}{\Delta} - m - \frac{F}{N}\right] \geq 0\text{pX} - \hat{\text{I}}^3\text{I} - m\left(\frac{\hat{\text{I}}^2\text{p} - \hat{\text{I}}^3\text{q}}{\beta\Delta}\right) - \frac{\gamma}{\beta}\frac{F}{N} \geq$$

When if $\hat{\text{I}}^2 = \hat{\text{I}}^3$ the expression reduces to: $= pX - \hat{\text{I}}^3\text{I} - \frac{F}{N} - m$



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